

Explainable Ensemble Learning for Heart Disease Prediction Using Clinical Data

Abstract—This study presents an explainable ensemble-learning workflow for heart-disease prediction from routine clinical data. The primary dataset contains 918 records, 11 predictors, and a binary outcome. Seven classifiers are tuned by five-fold grid search on an 80/20 stratified split, followed by soft-voting and stacking ensembles. XGBoost achieves the best individual-model accuracy (89.67%) and F1-score (0.9091), whereas K-nearest neighbors gives the highest individual ROC-AUC (0.9514). Stacking reaches 89.67% accuracy and 0.9389 ROC-AUC. Isotonic calibration lowers the Brier score from 0.0853 to 0.0749, and entropy-based selective prediction retains 95.1% of cases with 92.0% accuracy on the retained subset. Permutation importance and SHAP identify six influential variables, including ST-segment slope and chest-pain type. A compact six-feature stacking model records 86.96% accuracy, 0.8835 F1-score, and 0.9253 ROC-AUC. The study also reports a secondary-cohort implementation check and discusses validation limitations that affect clinical interpretation.

Index Terms—heart disease prediction, ensemble learning, XGBoost, stacking, probability calibration, SHAP, uncertainty estimation, clinical machine learning

I. INTRODUCTION

Cardiovascular risk depends on interacting clinical measurements such as age, cholesterol, blood pressure, chest-pain characteristics, exercise response, and electrocardiographic findings. Recent studies show that machine-learning models can capture these interactions across tabular, ensemble, and decision-support settings [1]–[3], [7], [9]. However, a clinically useful system should provide more than a class label. Its probabilities should be reliable, uncertain cases should be identifiable, and predictions should be explainable.

These requirements motivate probability calibration, selective prediction, and feature-attribution methods. Recent cardiovascular studies increasingly combine ensemble learning with explainable-AI tools such as SHAP and LIME [8], [12], [13], [15]. Careful validation is equally important because dataset choice, preprocessing, and evaluation design strongly affect reported performance [5], [11], [14].

This paper documents an end-to-end workflow that compares seven classifiers and two ensembles, evaluates calibration and uncertainty-aware deferral, explains predictions with permutation importance and SHAP, and retrains a compact six-feature model. It also audits the implementation so that the reported results are interpreted within the actual experimental design.

TABLE I
SUMMARY OF THE TWO SUPPLIED CLINICAL DATASETS

Item	Primary dataset	Secondary Cleveland dataset
File	heart.csv	heart_cleveland_upload.csv
Records	918	297
Predictors	11	13
Target	HeartDisease	condition
Positive	508 (55.34%)	137 (46.13%)
Negative	410 (44.66%)	160 (53.87%)
Role in code	Main model development and reported internal metrics	Second-dataset evaluation after model locking

II. RELATED WORK

Recent comparisons evaluate logistic regression alongside tree-based models, support-vector machines, nearest-neighbor methods, and boosting for heart-disease prediction [2], [3], [9], [10]. Ensemble systems combine complementary learners through voting or a meta-classifier and often improve robustness over a single model [4], [6], [13]. Explainable extensions use SHAP or LIME to provide global and patient-level interpretation [8], [12], [15]. Multi-source studies also highlight the importance of dataset integration and independent evaluation [16]. The present workflow additionally examines calibration and uncertainty-based deferral instead of treating discrimination alone as sufficient.

III. MATERIALS AND METHODS

A. Data and Audit

The primary file, `heart.csv`, contains 918 observations, 11 clinical predictors, and the binary target `HeartDisease`; 508 cases are positive and 410 are negative. A recent multi-source study reports the same 918-record consolidation after duplicate removal [16]. The secondary Cleveland-formatted file contains 297 rows, 13 predictors, and 137 positive cases; recent comparative work likewise uses the Cleveland schema for heart-disease prediction [14]. Table I summarizes their roles.

The initial audit finds no explicit missing cells or additional duplicate rows. It flags one zero `RestingBP` value and 172 zero `Cholesterol` values as unavailable, then creates a 70/15/15 exploratory split. The metric-producing branch later reloads the raw file and creates a separate 80/20 stratified split (734 training and 184 evaluation records) with `random_state=42`.

Numerical predictors are processed with a five-neighbor KNN imputer and standard scaling; categorical predictors are one-hot encoded. Fig. 1 distinguishes the exploratory and metric-producing branches.

B. Models and Ensembles

Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, K-Nearest Neighbors, Gradient Boosting, and XGBoost are tuned by five-fold stratified GridSearchCV using ROC-AUC. Each complete preprocessing-and-model pipeline is evaluated by accuracy, precision, recall, F1-score, and ROC-AUC. Soft voting and stacking use the tuned Random Forest, Gradient Boosting, XGBoost, and Logistic Regression pipelines. Stacking uses Logistic Regression as the meta-classifier. If $p_m(x)$ is the positive-class probability from model m , soft voting is

$$p_{\text{vote}}(x) = \frac{1}{M} \sum_{m=1}^M p_m(x), \quad (1)$$

where $M = 4$.

C. Calibration, Uncertainty, and Explanation

The stacking model is calibrated with isotonic and sigmoid methods in `cv='prefit'` mode. Probability quality is measured by log loss and the Brier score,

$$\text{BS} = \frac{1}{N} \sum_{i=1}^N (p_i - y_i)^2. \quad (2)$$

For selective prediction, binary entropy measures uncertainty,

$$H(p) = -p \log_2 p - (1 - p) \log_2 (1 - p). \quad (3)$$

and cases above the selected threshold of 0.85 are deferred. Permutation importance ranks original variables by the reduction in ROC-AUC after shuffling. SHAP TreeExplainer is applied to the XGBoost base learner, so its plots explain that component rather than the full stack. The top-six experiment retains ST Slope, ChestPainType, Cholesterol, Oldpeak, Sex, and FastingBS, then retrain and sigmoid-calibrates the stacking model. Finally, the locked model is applied to the Cleveland-formatted file using the mapping executed in the notebook.

IV. RESULTS

A. Classifier and Ensemble Performance

Table II shows that XGBoost obtains the highest individual-model accuracy (89.67%) and F1-score (0.9091), while KNN achieves the highest individual ROC-AUC (0.9514). Logistic Regression and SVM both reach 0.9314 recall, indicating that simpler models remain competitive.

XGBoost's confusion matrix contains 70 true negatives, 12 false positives, 7 false negatives, and 95 true positives. Fig. 2 places it beside the representative ROC curves.

Both ensembles reach 0.8967 accuracy and 0.9082 F1-score (Table III). Stacking gives a slightly higher ROC-AUC than soft voting and is therefore used for calibration and uncertainty analysis.

B. Calibration and Selective Prediction

Isotonic calibration produces the strongest reported probability metrics (Table IV): the Brier score decreases from 0.0853 to 0.0749, log loss from 0.3004 to 0.2572, and measured ROC-AUC rises to 0.9504. Sigmoid calibration provides only a small Brier-score improvement.

At the entropy threshold of 0.85, the system accepts 175 of 184 cases (95.1% coverage), defers 9, and reaches 92.0% accuracy on accepted cases. Fig. 3 summarizes calibration and the coverage-accuracy trade-off.

C. Explainability Analysis

Permutation importance ranks ST Slope first, with a mean ROC-AUC decrease of 0.1055 after shuffling. ChestPainType, Cholesterol, Oldpeak, Sex, and FastingBS follow. The XGBoost SHAP summary independently emphasizes transformed ST-slope and chest-pain features (Fig. 4).

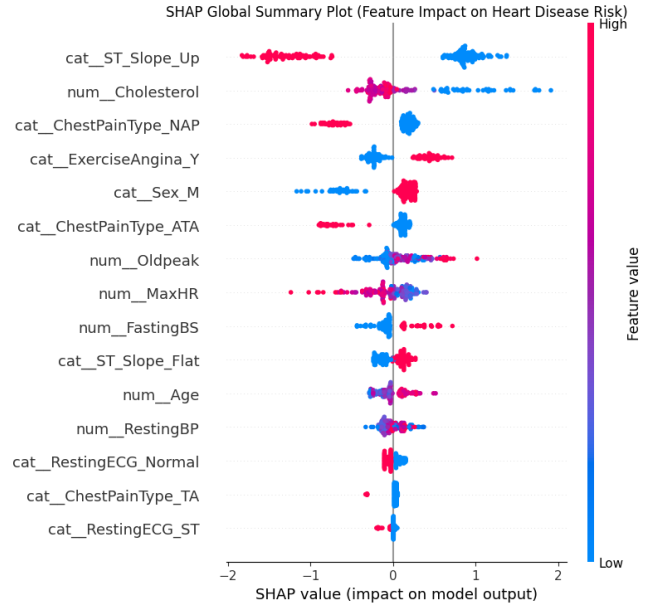


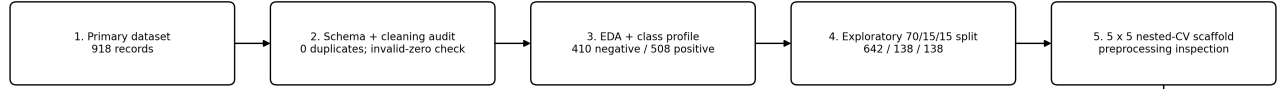
Fig. 4. Global SHAP summary for the XGBoost base learner. The permutation-importance ranking is reported in the accompanying text.

The global SHAP plot and permutation scores show which variables matter across the evaluation set. Because SHAP is applied to XGBoost rather than the complete stack, these attributions are interpreted as explanations of that base learner only.

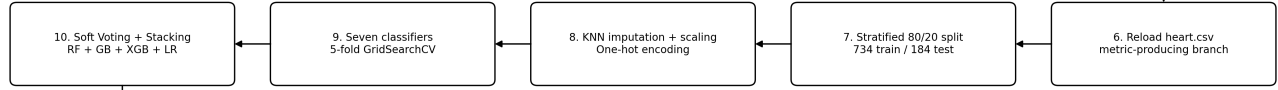
D. Feature Reduction and Final Six-Feature Model

Performance improves rapidly with the first several ranked features and then approaches a plateau. The selected six-feature model achieves 0.8696 accuracy, 0.8835 F1-score, 0.9253 ROC-AUC, and a 0.0990 Brier score (Table V). Its

Audit / exploratory branch (Steps 1-11)



Metric-producing modeling branch (Steps 12-18)



Calibration, explanation, final model and second-dataset check

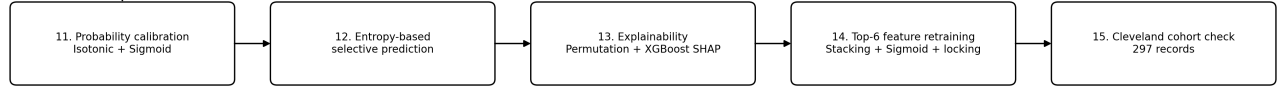


Fig. 1. Code-faithful methodology. Steps 1–11 perform data audit, exploratory splitting, and a nested-CV preprocessing scaffold. Step 12 reloads the primary CSV and creates the 80/20 split used by the reported model metrics. The final locked six-feature model is then applied to the supplied Cleveland-formatted dataset.

TABLE II
PERFORMANCE OF THE SEVEN TUNED INDIVIDUAL MODELS ON THE NOTEBOOK’S 20% TEST PARTITION

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.8859	0.8716	0.9314	0.9005	0.9308
Decision Tree	0.8152	0.8542	0.8039	0.8283	0.8947
Random Forest	0.8750	0.8835	0.8922	0.8878	0.9296
Support Vector Machine	0.8750	0.8559	0.9314	0.8920	0.9305
K-Nearest Neighbors	0.8859	0.8932	0.9020	0.8976	0.9514
Gradient Boosting	0.8913	0.8942	0.9118	0.9029	0.9372
XGBoost	0.8967	0.8879	0.9314	0.9091	0.9375

TABLE III
ENSEMBLE PERFORMANCE

Model	Accuracy	F1	ROC-AUC
Soft Voting	0.8967	0.9082	0.9385
Stacking	0.8967	0.9082	0.9389

TABLE IV
PROBABILITY CALIBRATION RESULTS

Method	Brier	Log loss	ROC-AUC
Uncalibrated stacking	0.0853	0.3004	0.9389
Isotonic	0.0749	0.2572	0.9504
Sigmoid	0.0845	0.2965	0.9389

TABLE V
FINAL SIX-FEATURE MODEL ON THE 20% INTERNAL PARTITION

Metric	Value
Accuracy	0.8696
Precision	0.8750
Recall	0.8922
F1-score	0.8835
ROC-AUC	0.9253
Brier score	0.0990

TABLE VI
INTERNAL VERSUS SECONDARY-COHORT METRICS

Metric	Internal	Secondary	Abs. diff.
Accuracy	0.8696	0.7104	0.1591
Precision	0.8750	0.6759	0.1991
Recall	0.8922	0.7153	0.1768
F1-score	0.8835	0.6950	0.1885
ROC-AUC	0.9253	0.7680	0.1573
Brier score	0.0990	0.2075	0.1085

confusion matrix contains 69 true negatives, 13 false positives, 11 false negatives, and 91 true positives.

E. Secondary Cleveland-Cohort Results

On the 297 Cleveland-formatted rows, the executed mapping gives 0.7104 accuracy, 0.6950 F1-score, 0.7680 ROC-AUC, and a 0.2075 Brier score. The corresponding confusion counts are 113 true negatives, 47 false positives, 39 false

negatives, and 98 true positives. Table VI shows the decline relative to the internal partition.

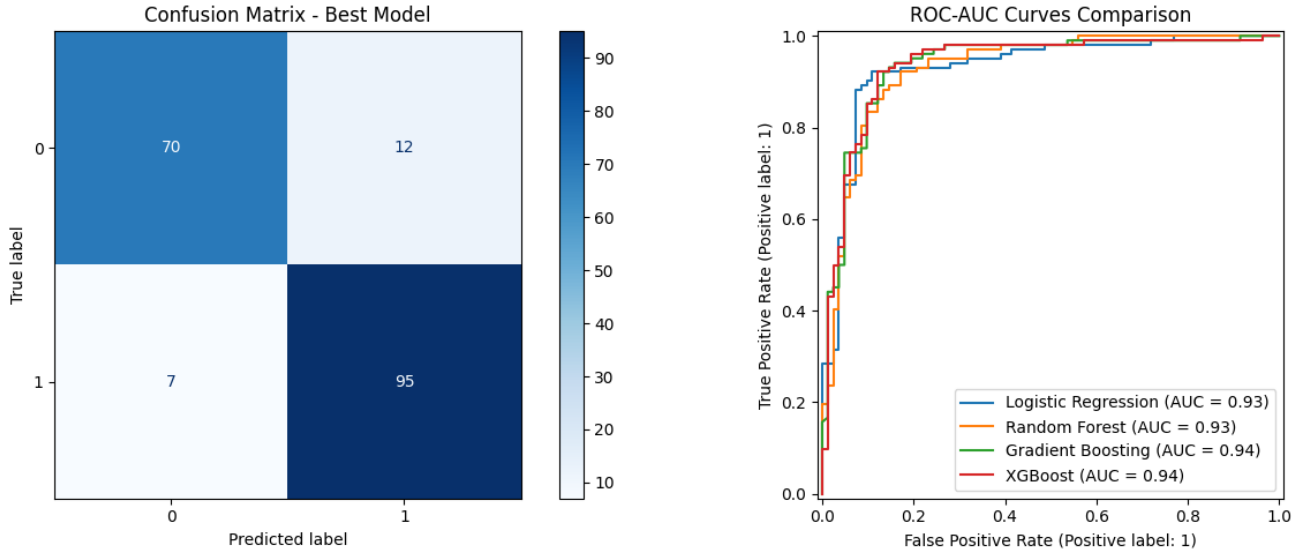


Fig. 2. Notebook output for the selected XGBoost model and representative ROC curves. Left: XGBoost confusion matrix. Right: ROC comparison of four tuned models.

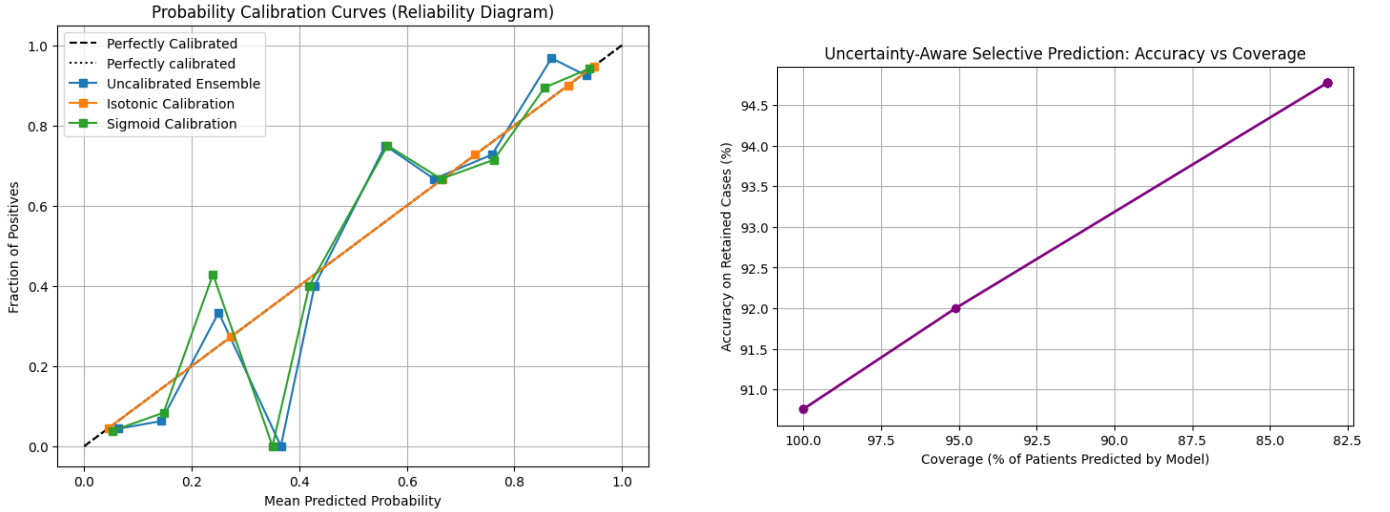


Fig. 3. Probability calibration (left) and entropy-based selective prediction (right). Higher coverage includes more uncertain cases and reduces retained-case accuracy.

V. DISCUSSION

The results show that model quality depends on the metric being emphasized. XGBoost leads accuracy and F1-score, while KNN gives the strongest individual ROC-AUC. Stacking does not improve accuracy over XGBoost, but it provides a consistent base for calibration, uncertainty estimation, and explanation.

Isotonic calibration improves the reported probability metrics, and selective prediction demonstrates a practical alternative to forcing decisions on uncertain cases. These findings are promising but exploratory because calibration and threshold selection use the same evaluation partition. Permutation importance and XGBoost SHAP broadly agree on ST-slope and chest-pain information, although SHAP explains only the

XGBoost component. Finally, the six-feature model preserves much of the ranking performance while reducing clinical input requirements; its lower ROC-AUC than the 11-feature point reflects a deliberate simplicity–performance trade-off.

VI. LIMITATIONS AND IMPLEMENTATION AUDIT

The reported metrics come from the later 80/20 branch, not the earlier 70/15/15 exploratory split, and the initial zero-value cleaning is not reused after the raw file is reloaded. Calibration is fitted and evaluated on the same 184 records, so its gains may be optimistic. In addition, the executed Cleveland mapping sets `FastingBS=0`, and source auditing shows that the 297 Cleveland rows overlap with the aggregate primary dataset. The secondary analysis is therefore an implementation check rather than an independent external validation. Future

work should use separate calibration and test sets, verified schema mapping, nested model selection, and provenance-disjoint cohorts.

VII. CONCLUSION

The proposed workflow combines competitive heart-disease prediction with calibration, uncertainty-aware deferral, and explanation. XGBoost is the strongest individual model by accuracy and F1-score, while stacking supports the later analysis stages. The six-feature model offers a practical compact alternative, but independent cohorts and stricter calibration design are required before clinical conclusions can be drawn.

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